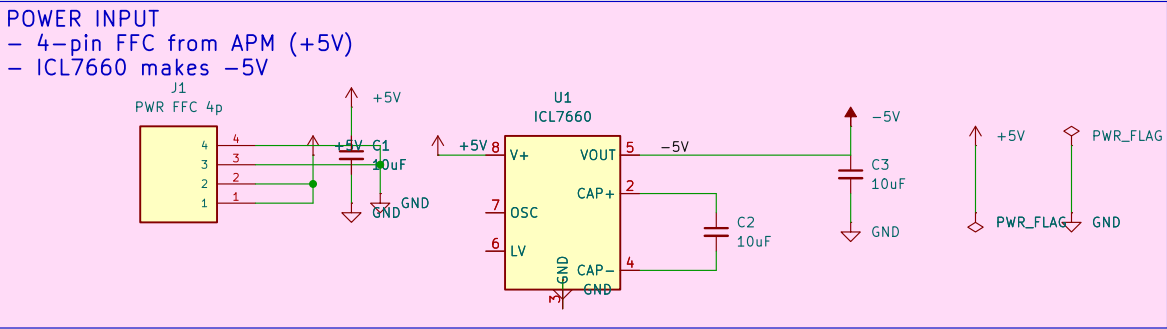
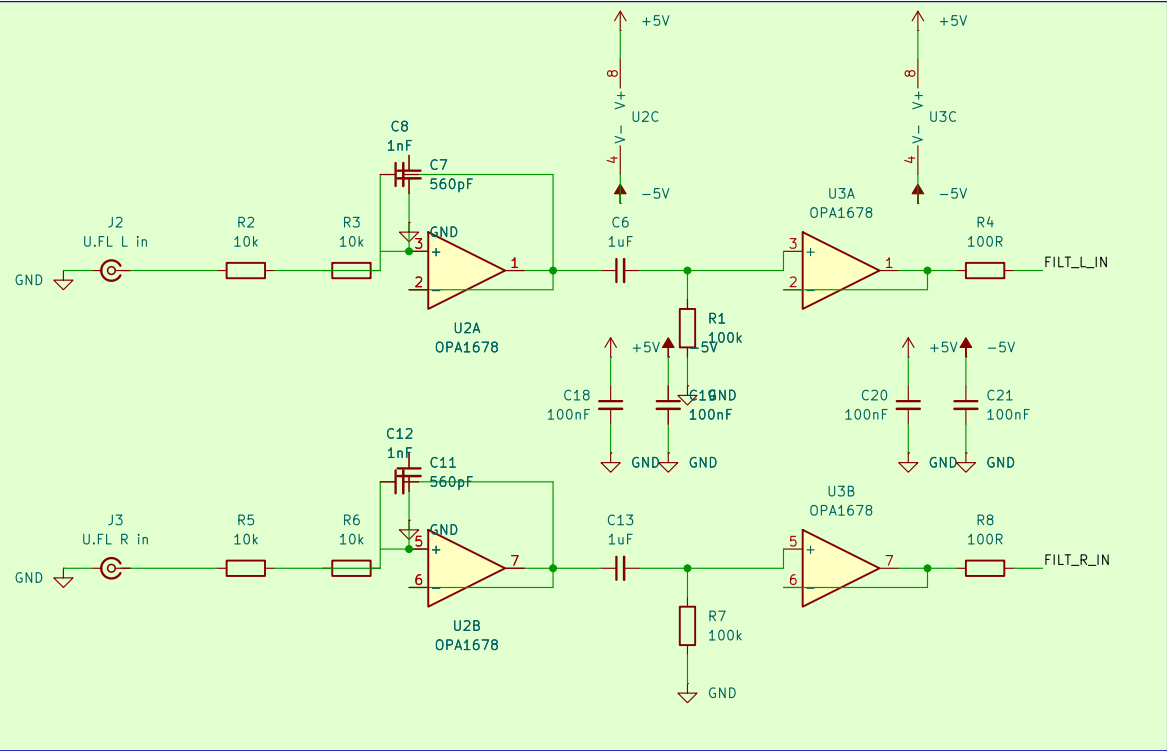


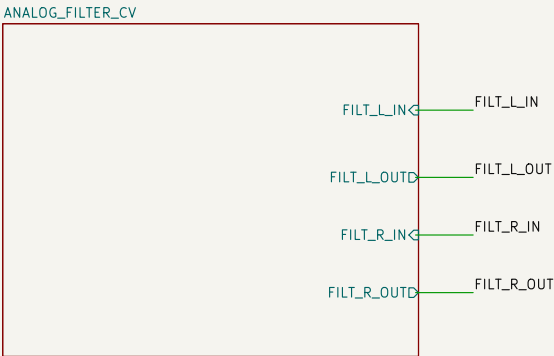
AUDIO BOARD – front end: power, U.FL audio in, reconstruction filter + buffers, 1/4in L/R + 3.5mm out



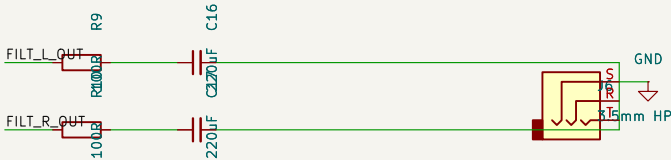
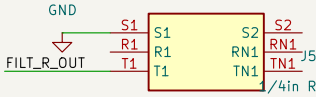
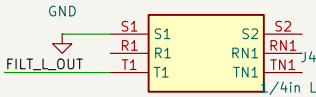
OUTPUTS – 1/4in L/R (ACJS-MHOM) + 3.5mm headphone/debug



ANALOG PATH – each channel's buffer output (FILT_x_IN) goes into the analog sub-sheet; FILT_x_OUT returns to the jacks



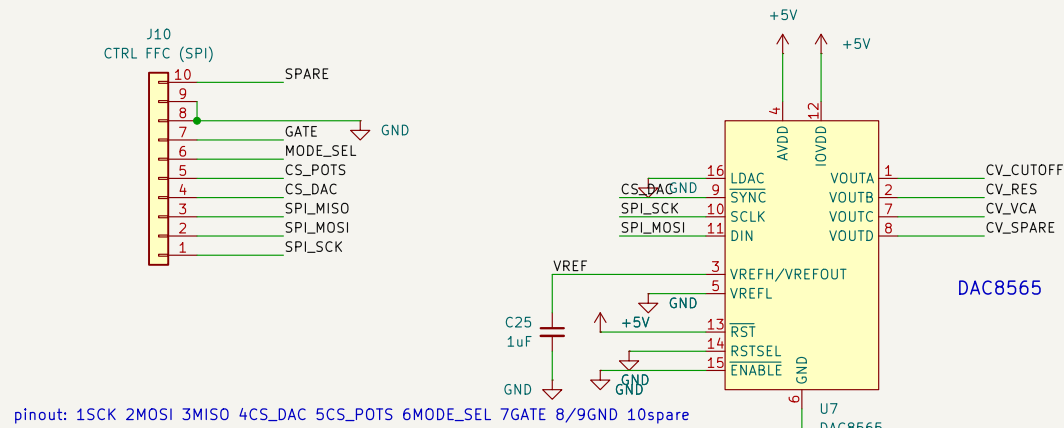
File: ANALOG_FILTER_CV.kicad_sch



ANALOG FILTER / CV (stereo) - SPI in (10-pin FFC) -> DAC8565 makes CVs -> per-channel LM13700 VCF/VCA; CD4053 selects analog or FPGA-digital

CONTROL - 10-pin FFC (SPI from APM)

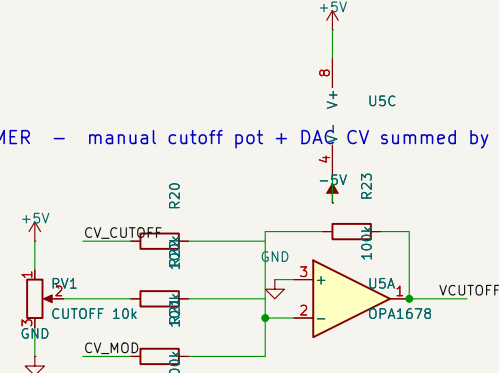
TOGGLE - CD4053: MODE_SEL selects analog (VCA_x_OUT) vs bypass (FILT_x_IN) for BOTH L and R



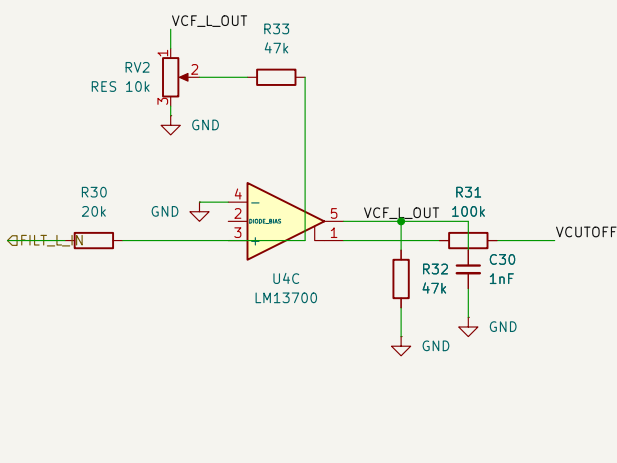
DAC8565 - 4-ch SPI CV DAC (cutoff / res / VCA voltages, on-board)

pinout: 1SCK 2MOSI 3MISO 4CS_DAC 5CS_POTS 6MODE_SEL 7GATE 8/9GND 10spare

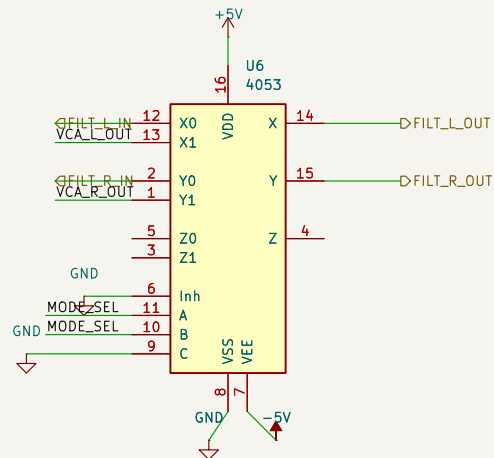
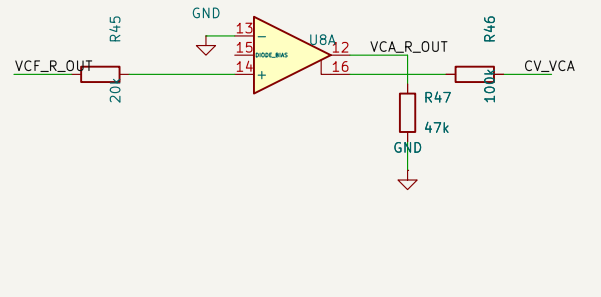
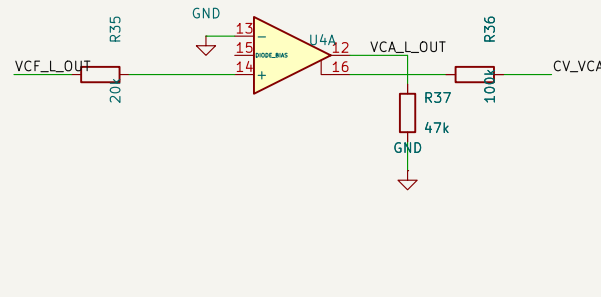
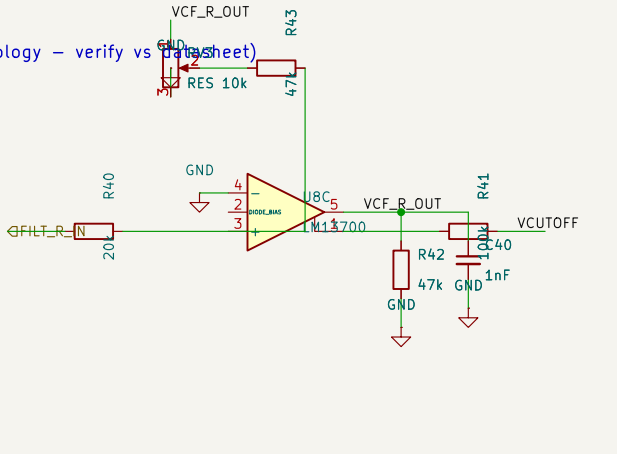
CV SUMMER - manual cutoff pot + DAC CV summed by U5 -> VCUTOFF (drives both L and R VCFs)



VCF+VCA L - LM13700 OTA lowpass then OTA VCA (NOTE: OTA/labc topology - verify vs datasheet)



VCF+VCA R - LM13700 OTA lowpass then OTA VCA (NOTE: OTA/labc topology - verify vs datasheet)



FILT_x_IN / FILT_x_OUT are the hierarchical pins shared with the root (front-end) sheet. Power rails (+5V/-5V/GND) are global.

Sheet: /ANALOG_FILTER_CV/ File: ANALOG_FILTER_CV.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 2/2